

Evaluation Datasets:

Transcripts:

<https://drive.google.com/file/d/1A4FbPbrpHfnColffC1RoKeitPbnGCd88/view?usp=sharing>

RNA Seq Queries:

https://drive.google.com/file/d/1lanLVsodNh4A9Pg_JRPSgkTnRx9qqPq9/view?usp=sharing

Baseline CPU System (Also applicable to Performance-Area Tradeoff Calculations in Report for Bonus points):

Intel(R) Xeon(R) CPU E5-2687W v4 @ 3.00GHz

128 GB DDR4

Baseline Performance (# Logic Units for Distribution = 256, k = 15)

Distribution on CPUs (/evaluation/distribute.cpp: distribute_default()) : 438.05 ms (1 Thread)

Indexing on CPUs (baseline/index_and_quantify.cpp: index_kmers()): 666.53 s (24 Threads)

Indexing on Single Logic Unit with largest workload (evaluation/index.c: generate_index_default()): *To be Announced using reduced data size* (1 Threads)

Quantification on CPUs (baseline/index_and_quantify.cpp: quantify()): 3.02 s (24 Threads)

Overall Runtime (For Ranking purposes involving 30 points): 669.55 s

NOTE:

1. A smaller number of inputs will be used for evaluating Indexing on Logic Units. For now, you can use a value of 10 or lower for transcript_sequences_count in transcripts_data.h and 100 in query_data.h
2. 256 Logic Units instead of the default 64 will be used for evaluations. So, please change 64 to 256 in Makefile for run_distribute if you are using the datasets provided for evaluations.
3. Your Indexing solution should have a lower runtime both for CPUs (666.53 s) and a single Logic Unit with the largest workload (TBA).